Status	Finished
Started	Wednesday, 21 October 2026, 4:07 PM
Completed	Wednesday, 21 October 2026, 4:15 PM
Duration	8 mins 25 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is large and unsorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It can be applied only if the list is sorted ✓
c. It works on unsorted lists
d. It always starts from the beginning of the list

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. In the entire list
c. At the beginning of the list
d. In the right sublist

Question **4**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array

b. When the item is not the array at all

c. Item is the last element in the array or item is not there at all

d. When the item is somewhere in the middle of the array ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search

b. Hash search

c. Linear search ✓

d. Both (1) & (3)

Question **6**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element

b. The search stops

c. The middle element is chosen randomly

d. The lower middle element is chosen as the middle element ✗

Question **7**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n-1
- c. n/2
- d. log n

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By using a hash function
- c. By comparing each element sequentially
- d. By dividing the list length by two ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Linear search ✓
- b. Dynamic search
- c. Binary search
- d. Hashing

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓

b. There should be direct access to the middle element in any sublist

c. The list must be sorted

d. Number values should only be present

Number values should only be present

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O(n log n)

c. O(log n) ✓

d. O👎

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers

b. It requires the list to be sorted ✓

c. It is slower than linear search for small lists

d. It can only be applied to large lists

Question **13**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎
b. O(log n)
c. O(1) ✓
d. O(n log n)

Question **14**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 100
c. 500
d. 1000

Question **15**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the middle of the list
c. In the left sublist
d. In the right sublist ✓